

ReplicationPad3d#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.padding.ReplicationPad3d.html>

Description:

Pads the input tensor using replication of the input boundary. For N-dimensional padding, use `torch.nn.functional.pad()`. padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses (padding_left\text{padding_left}padding_left, padding_right\text{padding_right}padding_right, padding_top\text{padding_top}padding_top, padding_bottom\text{padding_bottom}padding_bottom, padding_front\text{padding_front}padding_front, padding_back\text{padding_back}padding_back) Note that the output dimensions must remain positive. Input: (N,C,Din,Hin,Win)(N, C, D_{in}, H_{in}, W_{in})(N,C,Din,Hin,Win) or (C,Din,Hin,Win)(C, D_{in}, H_{in}, W_{in})(C,Din,Hin,Win). Output: (N,C,Dout,Hout,Wout)(N, C, D_{out}, H_{out}, W_{out})(N,C,Dout,Hout,Wout) or (C,Dout,Hout,Wout)(C, D_{out}, H_{out}, W_{out})(C,Dout,Hout,Wout), where

Function Signature

```
class torch.nn.modules.padding.ReplicationPad3d(padding)
[source]#
```

Parameters

Shape:

Code Examples

```
>>> m = nn.ReplicationPad3d(3)
>>> input = torch.randn(16, 3, 8, 320, 480)
>>> output = m(input)
>>> # using different paddings for different sides
>>> m = nn.ReplicationPad3d((3, 3, 6, 6, 1, 1))
>>> output = m(input)
```

Detailed Documentation

Documentation

Pads the input tensor using replication of the input boundary.

For N-dimensional padding, use `torch.nn.functional.pad()`.

padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses (padding_left\text{padding_left}padding_left, padding_right\text{padding_right}padding_right, padding_top\text{padding_top}padding_top, padding_bottom\text{padding_bottom}padding_bottom, padding_front\text{padding_front}padding_front, padding_back\text{padding_back}padding_back) Note that the output dimensions must remain positive.

Input: (N,C,Din,Hin,Win)(N, C, D_{in}, H_{in}, W_{in})(N,C,Din,Hin,Win) or (C,Din,Hin,Win)(C, D_{in}, H_{in}, W_{in})(C,Din,Hin,Win).

Output: (N,C,Dout,Hout,Wout)(N, C, D_{out}, H_{out}, W_{out})(N,C,Dout,Hout,Wout) or (C,Dout,Hout,Wout)(C, D_{out}, H_{out}, W_{out})(C,Dout,Hout,Wout), where

$D_{out} = D_{in} + \text{padding_front} + \text{padding_back}$
 $D_{out} = D_{in} + \text{padding_front} + \text{padding_back}$

$H_{out} = H_{in} + \text{padding_top} + \text{padding_bottom}$
 $H_{out} = H_{in} + \text{padding_top} + \text{padding_bottom}$

$W_{out} = W_{in} + \text{padding_left} + \text{padding_right}$
 $W_{out} = W_{in} + \text{padding_left} + \text{padding_right}$